

COLLIDING GALAXIES

THE RELATIVELY SHORT DISTANCES BETWEEN SOME GALAXIES COMPARED TO THEIR SIZE MAKES COLLISIONS FAIRLY COMMON. THESE SPECTACULAR EVENTS TRIGGER HUGE WAVES OF STAR FORMATION AND PLAY A KEY ROLE IN THE STORY OF GALAXY EVOLUTION.

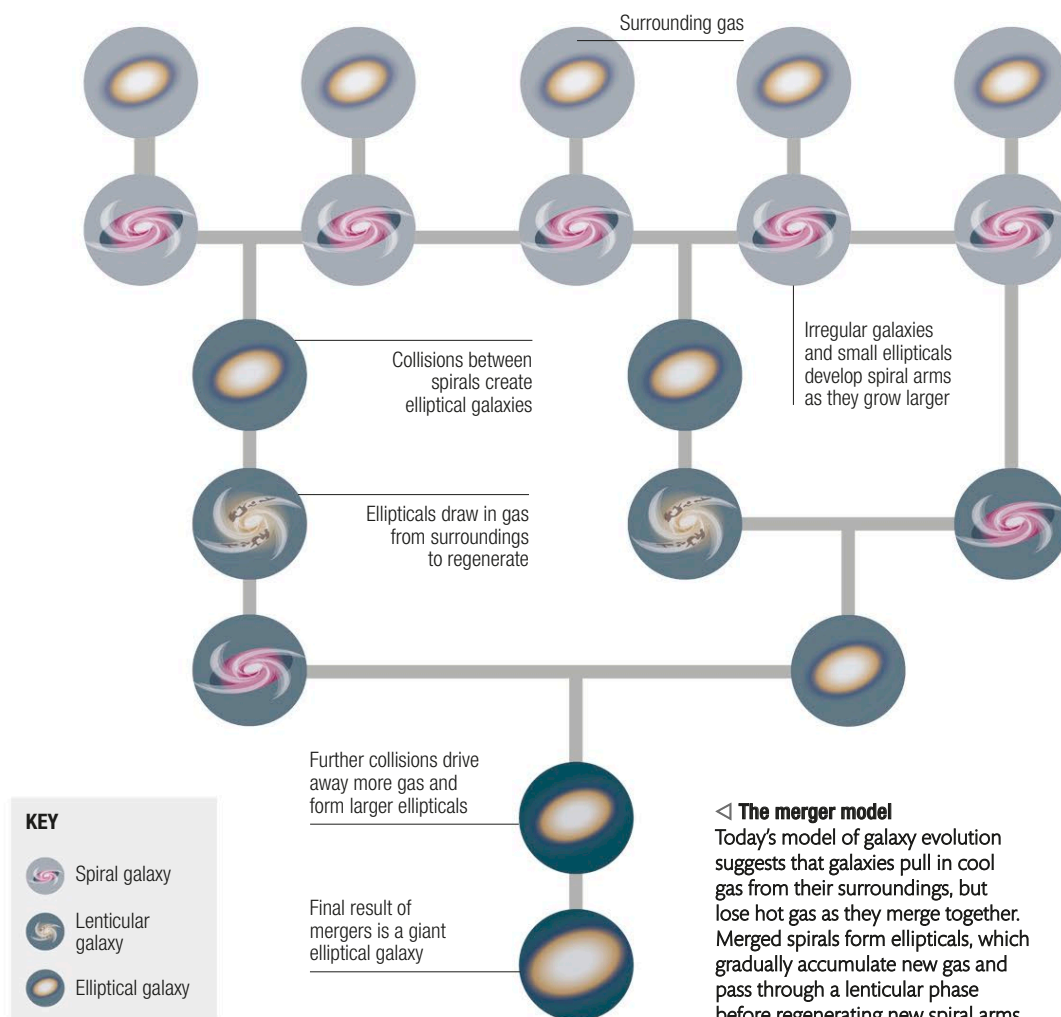
When galaxies collide, the stars within them are so widely spaced that they rarely hit one another. However, huge clouds of star-forming gas smash together in head-on collisions that compress and trigger vast new waves of star formation. The powerful gravity of the coalescing gas pulls the stars back toward the center, causing the colliding galaxies to merge over hundreds of millions of years.

Galaxy evolution

There is evidence that galaxies change from one type to another over time—irregular galaxies were far more common in the early Universe, spirals dominate today, and elliptical galaxies are most common in the heart of

galaxy clusters (see pp.66–67). As a result, astronomers think that collisions play a key role in galaxy evolution: gas-rich spirals initially formed from smaller irregular galaxies, while ellipticals are created by collisions between spirals, which send their stars into chaotic orbits, trigger vast waves of new star formation, and ultimately drive gas away into intergalactic space, where it can no longer form new stars.

Depending on the energy of the collision and the nearby environment, the merged galaxy's gravity may be able to pull back sufficient material from its surroundings to generate a new disk, and eventually to restart star formation, creating a new spiral that will eventually become part of another merger.



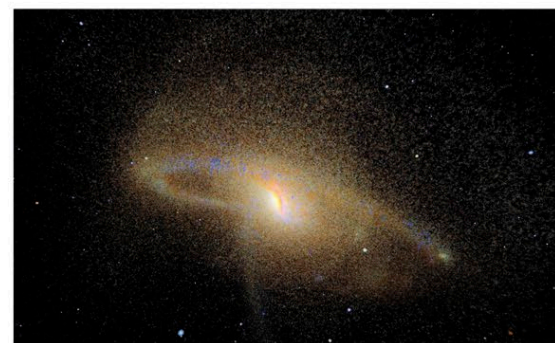
△ Close encounter

Near misses between galaxies are even more common than direct collisions. During these events, tidal forces are created that strengthen features such as spiral arms.



△ Unwinding arms

As galaxies come together, the orbits of stars within them are disrupted. Spiral arms unwind and their individual stars are scattered into intergalactic space.



△ Starburst

Most stars end up in chaotic orbits while gas is rammed together in huge star-forming clouds. Anchored by supermassive black holes, the galactic cores merge together.



△ Elliptical ending

The heating effects of the collision drive gas away from the galaxy, choking off the burst of star formation and leaving an elliptical system dominated by fainter, longer-lived stars.